

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



Department of Computer Science and Engineering

Course No: CSE 2104

Course Title: Data Structures Lab

Assignment No: 02

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Question:

You are given a list of books, each with a title, author and publication year. Your task is to sort the books by the publication year in descending order (most recent first) and if two books have the same publication year, sort them lexicographically by author in ascending order. If both the publication year and author are the same, sort the books lexicographically by title in ascending order. Implement this sorting using the Selection Sort algorithm. The input consists of an integer N (the number of books), followed by N lines of book details, where each line contains the title, author and publication year. Output the books in the sorted order, each on a new line in the format: "Title, Author, Publication Year".

Solution:

To solve the question, the Algorithms used is Selection Sort.

The algorithm for selection sort is:

```
for i <- 0 to n-2
    minIndex <- i
    for j <- i+1 to n-1
        if A[j] < A[minIndex]
            minIndex <- j
    swap(A[i] with A[minIndex])
```

C++ Code:

```
#include<bits/stdc++.h>
```

```
using namespace std;
```

```
typedef struct{
```

```
    string Author;
```

```
    string Title;
```

```
    int year;
```

```
}Book;
```

```
void SelectionSort(vector<Book> &bookInfo){
```

```
    for(int i = 0; i < bookInfo.size() - 1; i++){
```

```
        int target = i;
```

```
        for(int j = i + 1; j < bookInfo.size(); j++){
```

```
            if(bookInfo[target].year < bookInfo[j].year){
```

```
                target = j;
```

```
            }
```

```
else if(bookInfo[target].year == bookInfo[j].year){  
    if(bookInfo[target].Author > bookInfo[j].Author){  
        target = j;  
    }  
}
```

```
else if(bookInfo[target].Author == bookInfo[j].Author){  
    if(bookInfo[target].Title > bookInfo[j].Title){  
        target = j;  
    }  
}  
}  
}
```

```
swap(bookInfo[i], bookInfo[target]);  
}  
}
```

```
int main(){
```

```
    vector <Book> bookInfo;
```

```
int n;

cin >> n;

cin.ignore();

string temp;

Book tempBook;


for(int i = 0; i < n; i++){


    getline(cin, temp);

    int comma1 = temp.find(',');

    int comma2 = temp.find(',', comma1 + 1);


    tempBook.Title = temp.substr(0, comma1);

    tempBook.Author = temp.substr(comma1 + 2, comma2 - comma1 - 2);

    stringstream year(temp.substr(comma2 + 2));

    year >> tempBook.year;


    bookInfo.push_back(tempBook);
}


SelectionSort(bookInfo);
```

```

for(int i = 0; i < n; i++){

    cout << bookInfo[i].Title << ", " << bookInfo[i].Author << ", " << bookInfo[i].year << endl;

}

return 0;

}

```

Sample Input

5	Becoming, Michelle Obama, 2018
Sapiens, Yuval Noah Harari, 2011	Educated, Tara Westover, 2018
21 Lessons for the 21st Century, Yuval Noah Harari, 2018	21 Lessons for the 21st Century, Yuval Noah Harari, 2018
Educated, Tara Westover, 2018	Money: Vintage Minis, Yuval Noah Harari, 2018
Becoming, Michelle Obama, 2018	Sapiens, Yuval Noah Harari, 2011
Money: Vintage Minis, Yuval Noah Harari, 2018	

Sample Output